Fixed-chart progressive complete-owner assembly and anticipative-heat boundary

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1 Purpose and scope

R-103 closes complete H_N and REG only for the regular annular mutually orthogonal strict-past no-revisit class. The live question is what part of that conclusion survives for bounded smooth cylindrical-simple progressive controls when one physical covariance range is used more than once.

This note separates two statements which older roadmap prose sometimes called H_A simultaneously.

1. The *assembly statement* says that the complete R-079 endpoint and R-091 terminal nonduplication can be reconstructed on each fixed finite temporally faithful source chart, after which the exact R-083 and R-100–R-102 algebra gives the R-103 owner nomenclature and incidence template with no omitted cross term or duplicate owner. We denote the resulting algebraic defect by H_A^{asm}.
2. The *analytic statement* says that the resulting source action is bounded below uniformly in the cutoff, time mesh, and revisit multiplicity. R-093 identifies this statement with $\mathrm{OVERLAP}_{\mathrm{src}}$ and hence with the $q = 10/9$ Nelson bound.

The first statement is proved here. The second is not. This distinction is not cosmetic: the complete endpoint temporalises exactly, whereas the seven R-103 NEAR estimates and the eighth Cartan FAR estimate do not temporalise visit by visit.

The scope is fixed finite cutoff and fixed positive density floor, with the A1 production frame, common bounded entrywise-real-even local regulators, the R-079 canonical complete packet, and bounded smooth cylindrical-simple progressive controls. The temporal source chart must retain every past Brownian coordinate used by the controller, including kernel and auxiliary coordinates forgotten by the terminal physical map. All identities are finite-dimensional before any cutoff limit.

The note also records the exact heat measurability boundary. Deterministic or strict-past predictable heat may be frozen before the fresh root is integrated. A positive-semidefinite heat depending on that same fresh root cannot be inserted into the deterministic-heat argument by conditioning. An exact Gaussian second-chaos fixture detects the missing covariance.

No uniform $\mathrm{OVERLAP}_{\mathrm{src}}$ lower bound, Nelson estimate, cutoff or floor removal, interacting measure, or Sector-A closure is claimed. Tier stays T4.

2 Finite temporal source factorisation

Fix a finite time partition

$$0 = t_0 < t_1 < \cdots < t_K = 1, \qquad I_k = (t_{k-1}, t_k]. \tag{2.1}$$

Let the control be $u_t = u_k$ on I_k, where u_k is bounded, smooth cylindrical, and $\mathcal{F}_{t_{k-1}}$-measurable. In the finite terminal Gaussian space H_J, set

$$\Delta C_k = \int_{I_k} J_t J_t^* \, dt, \qquad L_k = \int_{I_k} J_t \, dt, \qquad a_k = L_k u_k. \tag{2.2}$$

No disjointness is assumed between the physical ranges $\operatorname{Ran} \Delta C_k$. They may overlap or coincide.

Theorem 2.1 (strict-past source factorisation and cost direction)

For every k there is a contraction coordinate R_k such that

$$L_k = \Delta C_k^{1/2} R_k, \qquad \|R_k\| \le |I_k|^{1/2}. \tag{2.3}$$

Consequently, with $S_k = \Delta C_k^{1/2}$ and $h_k = R_k u_k$,

$$\boxed{a_k = S_k h_k, \qquad \|h_k\|^2 \le |I_k| \|u_k\|^2.} \tag{2.4}$$

Proof. Operator Cauchy–Schwarz gives

$$L_k L_k^* \preceq |I_k| \Delta C_k. \tag{2.5}$$

Douglas range factorisation gives (2.3), including singular ΔC_k. Applying R_k to u_k proves (2.4). □

Let ξ_k be the independent standard source block for S_k. Add any kernel or auxiliary coordinates which the controller observes but the physical terminal map discards. Then h_k is a function only of earlier source and auxiliary blocks, and is therefore strictly triangular. The physical terminal field and shift are

$$X_J \overset{\text{law}}{=} S_\pi \xi := \sum_{k=1}^{K} S_k \xi_k, \qquad T_J u = \sum_{k=1}^{K} a_k = \sum_{k=1}^{K} S_k h_k. \tag{2.6}$$

Taking expectations in (2.4) yields

$$\boxed{\mathbb{E} \sum_{k=1}^{K} \|h_k\|^2 \le \mathbb{E} \int_0^1 \|u_t\|^2 dt.} \tag{2.7}$$

The direction in (2.7) is essential. It is a contraction from a progressive control to a source chart, not chartwise equality of the two costs.

3 Complete endpoint and owner temporalisation

Write the endpoint-translation path as

$$Z_k = S_\pi \xi + \sum_{r \le k} S_r h_r, \qquad Z_0 = S_\pi \xi. \tag{3.1}$$

Here $S_\pi \xi$ is the already fixed terminal Gaussian base. Thus Z_k is an algebraic endpoint-translation path, not an $\mathcal{F}_{t_k}$-adapted physical evolution. For every finite-dimensional endpoint functional F,

$$\sum_{k=1}^{K} \{F(Z_k) - F(Z_{k-1})\} = F(Z_K) - F(Z_0) \tag{3.2}$$

pathwise. The point of the R-079/R-091 algebra is that the same identity remains true after the production endpoint is expanded into its low, square, trace, heat, current, paid, and forest coordinates in expectation.

For one scalar current the cross term which is easiest to lose is

$$\frac{1}{2}\{(w+f+i)^2-w^2\}=wf+\frac{f^2}{2}+(w+f)i+\frac{i^2}{2}. \tag{3.3}$$

Replacing $(w+f)i$ by wi deletes fi. Under temporal refinement that deleted product is precisely a past/future or visit/revisit cross. Thus the complete square must be expanded only after its full terminal endpoint has been fixed.

Let $\mathfrak{P}^{\mathrm{comp}}_{J,\pi}$ denote the R-079 complete canonical packet on the temporal chart. On this fixed chart, let the superscript π mean that the exact R-083 spatial/production-row projections and the exact R-100/R-101/R-102 algebra are reapplied to the already reconstructed R-079 endpoint. R-103 supplies the owner names and incidence template only; none of its seven/eight quantitative regular estimates is used below.

Theorem 3.1 (exact fixed-chart complete-owner reconstruction)

For every fixed chart in the scope of Section 2,

$$\boxed{\begin{gathered}\langle\mathfrak{P}^{\mathrm{comp}}_{J,\pi}\rangle=\langle\mathfrak{B}_{\mathrm{low}}\rangle+\mathfrak{L}_\ell+\langle\mathfrak{C}^{\pi}_{\mathrm{far}}\rangle+\langle\Delta\mathcal{W}^{\pi}_{L,\Sigma}\rangle\\ +\langle\Delta\mathcal{W}^{\pi}_{R}\rangle+\langle\mathfrak{P}^{\pi}_{\mathrm{paid}}\rangle.\end{gathered}} \tag{3.4}$$

The rational endpoint has the exact three-owner split

$$\boxed{\Delta\mathcal{W}^{\pi}_{R}=\mathcal{R}^{\pi}_{Q}+\mathcal{M}^{\pi}_{U}+\mathcal{K}^{\pi}_{R}.} \tag{3.5}$$

For a matching row, payment, and baseline split,

$$\boxed{\mathcal{O}_{\mathcal{F}}(B,R)=\sum_a\mathcal{O}_{\mathcal{F}}(B_a,R_a).} \tag{3.6}$$

Equations (3.4)–(3.6) hold after the owner coordinates are recomputed on that fixed chart. If π' is a pure subdivision representation of the same terminal Gaussian field and the same terminal shift, with the filtration, reveal order, and admissible controls unchanged, then the recombined total on either side is unchanged. The labelled Doob and owner summands are recomputed on π' and are not asserted to agree term by term.

Here a *representation-preserving subdivision* means a coupled rewriting which leaves the terminal Gaussian random variable, terminal shift, original reveal order, and every sigma-field actually used by h unchanged, and admits no new controls. A split creating newly revealed independent coordinates is a new chart, not such a subdivision.

Proof. R-081 Theorem 8.1 puts every temporal displacement into a strict-past covariance-root block without requiring orthogonal physical ranges. Its complete-packet temporalisation retains the injected/future cross in (3.3). R-079 then gives the complete-current and complete-safe-packet identity on these blocks. R-091 equation (9.3) says that the low endpoint, paid subtraction, future insertion, and terminal endpoint occur once after the finite telescope. Applying the R-083 FAR/NEAR and production-row partitions to that already reconstructed endpoint gives (3.4).

The exact R-101 rational endpoint identity gives (3.5): the fixed-low raw-Wick difference stays inside the low owners, only the future full-Wick reveal belongs to $\mathcal{R}^{\pi}_{Q}$, current families three through five form $\mathcal{M}^{\pi}_{U}$, and the complete shifted current forms $\mathcal{K}^{\pi}_{R}$. R-100 row additivity gives (3.6); its positive posterior row gap and negative complete-square gap cancel and cannot be retained as a second reserve.

For a representation-preserving subdivision, intermediate endpoints inserted into (3.2) cancel before expectation. Exact Doob polarisation may exchange conditional covariance with posterior-mean mass. Both recomputations equal the same reconstructed terminal R-079 endpoint; R-100 separately gives exact row additivity inside each recomputation. No invariance is claimed after adding fresh roots, changing reveal order, or enlarging the admissible-control class. ☐

The theorem is an expectation-level owner identity. The endpoint telescope (3.2) is pathwise; the Wick–Doob cancellations used to identify individual owners are expectation identities. No pointwise sign is asserted.

4 Owner incidence and refinement boundary

The R-103 incidence ledger is the algebraic template reapplied on each fixed temporal chart:

Coordinate	Multiplicity	Temporal rule
Conditional and complete low	one each	Keep both distinct endpoints.
Complete linear rows	one	Heat, trace, and forest remain internal.
Future raw-Wick residual	one	Fixed-low raw Wick stays in the low owners.
Unshifted rational current	one	Use only current families three through five.
Shifted rational current	one	Retain the complete future insertion and its terminal-square subcoordinate.
Paid difference	one	It is exact subtraction, not an added repair.
Cartan FAR	one in REG	Eighth module; absent from the seven-module NEAR packet.
q/r, Schur, matrix payment	zero extra	They are internal complete-owner coordinates.
R-063 forest	zero extra	It reconstructs the endpoint once.
R-076 base cubic, R-086 duplicate rows	zero extra	These are refunded in the active coordinate.

The first seven modules through the paid difference form the NEAR audit when the three complete linear rows are treated as their single accepted module. Cartan FAR is the eighth REG module. The terminal rational square is an internal positive subcoordinate of the shifted-current module, not a ninth top-level module. This table states multiplicities, not lower bounds. In particular, the two following operations are different.

1. A representation-preserving subdivision only inserts cancelling endpoints into (3.2), so the recombined endpoint is unchanged; its individual owner coordinates need not be.

2. A coarsening is causal only when every child control is measurable at the left endpoint of the proposed coarse block. If a child observes noise created inside that block, the controls cannot be merged into one strict-past shift.

Nor can a causal orthogonal change of coordinates remove repeated physical ranges. A block-lower-triangular orthogonal map has a block-lower-triangular inverse, but its inverse is also its block-upper-triangular transpose. It is therefore block diagonal. This is the R-093 causal-QR boundary.

5 Why the R-103 estimates do not temporalise

Theorem 3.1 does not apply the seven R-103 NEAR estimates or the eighth REG estimate to each visit. Two exact fixtures show why.

First take a scalar current $J(a) = a^2$. Visit one physical range by t and later revisit it by $-t$, so the final displacement is zero. The isolated conditional-low loss may be

$$\mathfrak{L}_{\text{low}} = -\frac{t^4}{2}, \qquad X = 2t^2, \qquad Y = 0. \tag{5.1}$$

No fixed η, C makes $-t^4/2 \geq -2\eta t^2 - C$ for all t. The matching complete-low endpoint is $+t^4/2$ and cancels (5.1) exactly. Thus the complete owner survives while the visitwise low estimate is false.

Second, on an event E of probability p, let

$$h_1 = p^{-1/2}\mathbf{1}_E e, \qquad h_2 = -h_1, \qquad \phi = Ke \neq 0. \tag{5.2}$$

Then the total source cost is two and the final physical displacement is zero, but

$$\|Kh_1\|_6^6 + \|Kh_2\|_6^6 = 2\|\phi\|_6^6 p^{-2}. \tag{5.3}$$

Hence absolute sixth-moment summation over visits diverges as $p \downarrow 0$. This is the R-094 revisit obstruction in the present complete-owner coordinate.

Both fixtures leave the signed full endpoint intact. Neither is a counterexample to $\text{OVERLAP}_{\text{src}}$ or Nelson.

6 Exact predictable versus anticipative heat boundary

Let G be a standard real Gaussian and let

$$Q(G) = G^2 - 1. \tag{6.1}$$

If a nonnegative random heat Σ is measurable in a strict-past sigma field independent of G, then

$$\mathbb{E}[\Sigma Q(G)] = 0. \tag{6.2}$$

Thus a deterministic-heat identity may be disintegrated over genuinely predictable heat, provided the fresh-root conditional law and the required uniform constants are retained.

This argument fails for same-root heat. Take the positive-semidefinite choice

$$\Sigma(G) = G^2. \tag{6.3}$$

Then

$$\boxed{\mathbb{E}[\Sigma(G)Q(G)] = \mathbb{E}[G^4] - \mathbb{E}[G^2] = 3 - 1 = 2.} \tag{6.4}$$

The missing term is exactly a coefficient/heat–Wick covariance. It has no reason to vanish when the heat sees the same innovation.

Proposition 6.1 (conditioning boundary)

Uniform deterministic-PSD-heat identities disintegrate through a random heat when that heat is measurable before, and conditionally independent of, the fresh Gaussian root to which the identity is applied. There is no general extension to arbitrary control-dependent or anticipative heat, even in one dimension.

Proof. Equation (6.2) is conditional centering. Equations (6.3)–(6.4) give a positive-semidefinite same-root counterexample. ☐

This fires `NG-2026-07-28-A13-ANTICIPATIVE-RANDOM-HEAT-CONDITIONING`. It narrows the R-103 heat boundary without refuting its deterministic-heat theorem. Future heat generated by later adapted controls is not automatically predictable at the earlier root.

7 Closure of the algebraic assembly module

Put

$$G_J(\phi) = V_J^{\mathrm{ren}}(\phi) + \frac{3}{20}\|\phi\|_6^6, \qquad q = \frac{10}{9}. \tag{7.1}$$

For a temporally faithful triangular source chart define

$$\mathcal{A}_J(h) = \mathbb{E}G_J(S_\pi(\xi + h)) + \frac{9}{20}\mathbb{E}\sum_k \|h_k\|^2. \tag{7.2}$$

Define $\mathcal{V}_{J,\pi}^{\mathrm{own}}(h)$ to be the recombined right-hand side of (3.4), with the control-independent R-079 baseline restored, and define the endpoint/owner-coordinate defect by

$$H_A^{\mathrm{asm}}(J,\pi,h) := \mathbb{E}V_J^{\mathrm{ren}}(S_\pi(\xi + h)) - \mathcal{V}_{J,\pi}^{\mathrm{own}}(h). \tag{7.3}$$

Set

$$\mathcal{A}_{J,\pi}^{\mathrm{own}}(h) := \mathcal{V}_{J,\pi}^{\mathrm{own}}(h) + \frac{3}{20}\mathbb{E}\|S_\pi(\xi + h)\|_6^6 + \frac{9}{20}\mathbb{E}\sum_k \|h_k\|^2, \tag{7.4}$$

Theorem 7.1 (fixed-chart endpoint-owner identity and cost slack)

The complete R-079 packet on every finite bounded smooth cylindrical-simple progressive control admits a temporally faithful source realisation satisfying (2.6)–(2.7). On that realisation, equations (3.4)–(3.6) reconstruct the physical endpoint in (7.2) exactly with every owner in the R-103 incidence template active once or refunded zero times. Therefore the algebraic assembly defect is zero:

$$\boxed{H_A^{\mathrm{asm}}(J,\pi,h) = 0, \qquad \mathcal{A}_{J,\pi}^{\mathrm{own}}(h) = \mathcal{A}_J(h).} \tag{7.5}$$

Proof. The source realisation and cost direction are Theorem 2.1. The expectation- level endpoint and owner identity is Theorem 3.1. Restoring the fixed baseline proves the first equality in (7.5); adding the same stabiliser and source cost to both sides proves the second. □

For clarity, the original progressive action is

$$\mathcal{A}_J^{\mathrm{phys}}(u) := \mathbb{E}G_J(X_J + T_J u) + \frac{9}{20}\mathbb{E}\int_0^1 \|u_t\|^2 dt. \tag{7.6}$$

Equations (2.6)–(2.7) give the separate one-way comparison

$$\boxed{\mathcal{A}_J^{\mathrm{phys}}(u) = \mathcal{A}_J(h) + \mathfrak{D}_{\mathrm{CM}}(u,h), \quad \mathfrak{D}_{\mathrm{CM}} := \frac{9}{20}\mathbb{E}\left[\int_0^1 \|u_t\|^2 dt - \sum_k \|h_k\|^2\right] \geq 0.} \tag{7.7}$$

H_A^{asm} is local notation for the endpoint/owner defect (7.3). Equation (7.5) is an identity only: it contains no sign, coercivity, mesh- uniform estimate, or equality of source and original progressive costs. No assertion about any bare H_A gate or the older Gaussian quadratic notation $H_A(x) = \langle x, Ax\rangle$ is made.

8 The exact remaining analytic theorem

Let $\mathfrak{T}_J$ be the R-093 directed union over all temporally faithful finite source refinements. R-093 Theorem 6.2 gives

$$\boxed{\inf_{(S_\pi,h)\in\mathfrak{T}_J} \mathcal{A}_J(h) = -\frac{9}{10}\log\mathbb{E}\exp\{-\frac{10}{9}G_J(X_J)\}.} \tag{8.1}$$

The corresponding exact Gibbs-gap identity is

$$\mathcal{A}_J(h) = -\frac{9}{10}\log Z_J + \frac{9}{10}H(\nu_h \mid \nu_J^*) + \frac{9}{10}\Phi_h, \tag{8.2}$$

where

$$Z_J = \mathbb{E}\exp\{-\frac{10}{9}G_J(X_J)\}. \tag{8.3}$$

Both gap terms in (8.2) are nonnegative, but R-093 near minimisers make both tend to zero. They are not an independent uniform reserve.

Consequently the next assertion

$$\boxed{\inf_{(S_\pi,h)\in\mathfrak{T}_J} \mathcal{A}_J(h) \geq -C \quad \text{uniformly in } J} \tag{8.4}$$

is precisely $\mathrm{OVERLAP}_{\mathrm{src}}$ and, by (8.1), precisely the $q = 10/9$ Nelson estimate. Theorem 7.1 removes an assembly ambiguity but does not prove (8.4).

The strongest currently reusable progressive estimates remain the R-089 global terminal Cameron–Martin contraction, its Hilbert martingale ledgers, the pure-control quartic terminal bridge, and the R-099 chronological Doob–Hardy coordinate ledger. None controls the complete nonlinear coefficient/Jensen/low-prefix packet. The explicit residual must remain inside the signed complete owner until (8.4) is proved.

9 Attempt, failure, and evidence map

Route	Verdict	Exact reason
R-081 Douglas source factorisation	closed	Terminal displacement is exact and source cost contracts.
Complete packet temporalisation	closed	R-079/R-091 preserve the future cross and terminal endpoint.
Fixed-chart owner recomputation	closed algebraically	The recombined owner has the R-103 incidence template; summands need not be termwise invariant.
Apply seven/eight R-103 bounds per visit	failed	Opposite revisit has quartic isolated low loss against quadratic cost.
Absolute revisit sixth moments	failed	Equation (5.3) grows like p^{-2} at fixed source cost.
Replace overlap by terminal polar control	failed	Terminal covariance compression need not preserve causality or coefficient coupling.
Condition through arbitrary random heat	failed	Same-root PSD heat has covariance defect two in (6.4).
Use fibre entropy as uniform reserve	failed	R-093 near minimisers force the fibre gap to zero.
Uniform source action	open	By (8.1) it is the Nelson theorem itself.

The successful and failed routes are recorded in the append-only exploration ledger. Existing R-080, R-089, R-093, and R-094 negative authorities remain binding and are not duplicated under new names. Only the anticipative-heat conditioning failure receives a new negative-result entry.

10 Devil's-advocate review

1. **Objection: Theorem 7.1 proves the full progressive gate. UPHELD as false.** It proves the exact assembly defect is zero. The gate statement is the uniform inequality (8.4), which remains open.

2. **Objection: Exact temporalisation transfers the R-103 module estimates. UPHELD as false.** Equations (5.1)–(5.3) show that separate low and sixth-moment visit estimates fail even though the complete signed endpoint cancels.

3. **Objection: The cost in (2.7) is equal to the original control cost. VALID WITH CORRECTION.** Douglas gives a non-increasing cost. Kernel directions may be removed. Only the infimum over the temporally faithful union is identified by R-093.

4. **Objection: Refinement invariance permits arbitrary coarsening. UPHELD as false.** Coarsening loses strict-past measurability whenever a child control observes noise produced inside the coarse interval.

5. **Objection: Source-block orthogonality makes repeated physical ranges orthogonal. UPHELD as false.** Independent source coordinates can map to the same physical range. Causal orthogonal QR cannot merge them.

6. **Objection: R-103 deterministic heat automatically covers every random heat. UPHELD as false.** Genuinely predictable heat may be frozen before fresh-root integration, while equation (6.4) shows that no automatic extension to arbitrary anticipative heat is valid. This is a sufficient condition and a no-general-extension boundary, not a necessary-and-sufficient classification.

7. **Objection: The exact Gibbs gap supplies a positive budget for the remaining packet. UPHELD as false.** Both gap terms can vanish along variational near minimisers. Assuming a uniform reserve would contradict R-093 Corollary 6.3.

8. **Objection: The scalar fixtures disprove Nelson. DISMISSED.** They refute visitwise, anticipative-conditioning, or noncausal-compression proof routes. Every omitted complete owner is required before a production counterexample could be claimed.

Hardcode masking is rejected: source cost comes from event probability and amplitude, Douglas slack from the computed kernel-cost gap, and Gaussian moments from two algorithms. The fixtures are dimensionless and finite- cutoff; no convergence or removal claim is made.

External review should focus on the cost direction in (2.7), the expectation- level versus pathwise distinction in Theorem 3.1, the owner multiplicities, the causality condition for coarsening, and the firewall between (7.3) and (8.4).

11 Executable verification

The primary executable checks Douglas factorisation and its strict pure-kernel slack; endpoint, square, and expectation-only Doob telescopes; a rejected pathwise Doob mutant; noncausal covariance compression; the full eight-module R-103 owner/refund incidence, including Cartan FAR and the nested terminal square; revisit and heat fixtures; the $q = 10/9$ coefficients; and every closure firewall. Gaussian moments are generated by double factorials.

The independent executable uses standard-library fractions, exact finite source enumeration, rational telescopes, an independently ordered owner audit, and a separate Gaussian recurrence; it imports neither the primary route nor NumPy, SymPy, or SciPy. The integrated verifier reruns both, pins R-079, R-081, R-083, R-089, R-091, R-093, and R-099–R-103, audits every package surface, and rejects claims that $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson, or Sector A is closed.

Run from the repository root:

```
E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/
  a13_classii_lossless_progressive_complete_owner_assembly_heat_boundary_verify.py
```

Result footer

```
Result ID: A13-CLASSII-LOSSLESS-PROGRESSIVE-COMPLETE-OWNER-
           ASSEMBLY-HEAT-BOUNDARY
Ledger:    R-104.
Precise statement:
           Every fixed-cutoff bounded smooth cylindrical-simple progressive
           control has a temporally faithful strict-triangular source
           realisation with the same terminal displacement and no larger
           source cost. On each fixed chart the exact R-079/R-091 endpoint
           and R-083/R-100--R-102 owner algebra reconstruct the endpoint
           across overlapping physical ranges; R-103 supplies nomenclature
           and incidence only. Representation-preserving subdivisions keep
           only the recombined total, not the labelled owner summands. Thus
           the endpoint-owner defect H_A^asm is zero. The physical action is
           the source action plus the nonnegative Douglas cost slack.
           Predictable fresh-root-independent heat can be disintegrated;
           same-root PSD heat shows no arbitrary anticipative extension.
Scope:     Finite cutoff, fixed positive floor, A1 production frame, common
           real-even local regulators, finite temporally faithful source
           refinements, and bounded smooth cylindrical-simple progressive
           controls. Exact endpoint identities are pathwise; owner identities
           involving Wick--Doob cancellation are expectation-level.
Dependencies:
           R-079, R-081, R-083, R-089, R-091, R-093, R-099--R-103.
Evidence grade:
           ANALYTIC, EXACT, EXECUTED, ALGEBRAIC-ENDPOINT-IDENTITY.
Reproduction command:
           E:\Dev\TECT.venv\Scripts\python.exe
           codes/foundations/
           a13_classii_lossless_progressive_complete_owner_
           assembly_heat_boundary_verify.py
Expected output:
           93/93 + 66/66 + 290/290 = 449/449 PASS; exit zero.
Negative result:
           NG-2026-07-28-A13-ANTICIPATIVE-RANDOM-HEAT-CONDITIONING.
Falsification gate:
           Failure of Douglas factorisation/cost direction, terminal or
           square telescope, R-079/R-091 temporalisation, fixed-chart
           R-100--R-103 owner incidence, Douglas slack, predictable heat
           centering, same-root heat fixture,
           authority/hash/PDF/exploration/registry/count contracts, or scope
           firewalls.
Tier before / after:
           T4 / T4; no promotion.
No-overclaim statement:
           No visitwise extension of the seven/eight R-103 estimates,
           cutoff-uniform OVERLAP_src lower bound, q=10/9 Nelson estimate,
           cutoff/floor removal, interacting measure, T5--T7, or Sector-A
           closure is proved.
Next required action:
           Prove (8.4), equivalently q=10/9 Nelson; then removals and measure.
```